

CAREER & ATS AI PLATFORM

Project Overview

Intelligent career-preparation platform helping users optimize ATS resumes, match job descriptions, and practice viva questions via Google Gemini API.

Problem Statement:	Job applicants struggle to format resumes for ATS compliance and lack personalized feedback for technical rounds.
Solution Implemented:	Built an ATS optimization platform combining spaCy NLP keyword parsing with Google Gemini LLM mock interview generation.
My Technical Role:	Lead AI Engineer

Main Features & Technical Stack

Core Features:	Technologies Used:
• ATS resume scoring & missing keyword detection • Job description skill matching using NLP semantic embeddings • Gemini API dynamic interview question generation • Real-time answer evaluation & performance feedback	React, Vite, FastAPI, Python, Google Gemini API, spaCy, SQLite, Vercel
	Architecture Flow:
	React Client → FastAPI Backend → spaCy NLP Engine → Gemini API → Vercel Host

Technical Challenge & Results

Key Engineering Challenge: Built layout-aware PDF parsers to group text blocks correctly across multi-column resumes.

Validated Results: Boosted candidate ATS resume scoring match rates by over 40%.

Live Demo: <https://ai-coach-ochre-pi.vercel.app> | **GitHub:** <https://github.com/prashant-singh-78> | **Portfolio:** <prashant08.vercel.app>